



[The exam text is in English. You should answer the questions in English.]

The exam consists of 13 questions (8 pages). Please first check if any page is missing in the printout you. For the open questions: please keep the handwriting clear and readable.

The exam starts at 9.00 hrs and ends at 12.00 hrs. Participation in the exam requires being present for **at least 1 hour**.

Electronic devices are not allowed and must be stowed.

Points for each open question are indicated.

- Please write readable and write your name and student identification number on every single sheet of paper for unique identification! **Content written on other sheets than these will not be evaluated.**
- **Don't spend much time writing long stories – that's NOT necessary. Don't panic! You CAN make it!**
- Look at the whole exam first to get an overview, pay attention to the grading scheme of multiple-choice questions, and do those assignments first which you feel are doable for you.
- Good success and all the best for your studies!

Part I - Multiple-choice [50 points].

- Each problem has a variable number of correct statements. Please write your answers in the table given at the end of the question sheet.
- In NO problem, all statements are correct. If you choose all the options, i.e., "ABCD", the score for that question will be ZERO!
- The score for each question will be given according to Morgan method:

$$5 \times \max \{ r / \#correct\ option(s) - w / \#distractors(s), 0 \},$$

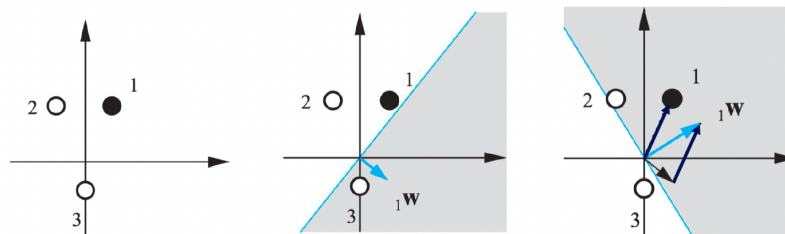
where:

- 5 is the total score for each question;
- r is the number of option(s) you select correctly;
- w is the number of option(s) you select incorrectly;
- $\#correct\ option(s)$ is the number of correct options of this question;
- $\#distractors(s)$ is the number of incorrect options/distractors of this question;
- Please DO NOT mark an option if you are not certain with it as a penalty will incur.

Please write the alphabetic characters (e.g., CD) representing your answers of the multiple-choice questions into the table below.

No.	1	2	3	4	5	6	7	8	9	10
Answers	ACD	ACD	A	CD	BD	B	C	ABD	AD	BC

1. In the McCulloch-Pitts model, what do synaptic weights represent? [more than one answer is correct]
 - a. Strength of the connection between two neurons.
 - b. The direction of ion flow.
 - c. Whether an input is excitatory or inhibitory.
 - d. A parameter determining the impact of an input on the neuron's output.
2. Which of the following statements about Hopfield Networks as associative memory systems are correct? [more than one answer is correct]
 - a. Stored patterns act as attractors in the network's state space.
 - b. They require supervised learning with labelled data.
 - c. They can handle incomplete or noisy input patterns.
 - d. They retrieve stored patterns based on similarity to input patterns.
3. Which of the following statements is wrong about Hebbian Learning? [one answer is correct]
 - a. Hebbian learning guarantees that all stored patterns in a Hopfield network can be perfectly retrieved without error, regardless of the number of patterns.
 - b. Hebbian learning can be summarized by the phrase "neurons that fire together, wire together."
 - c. Hebbian Learning provides a simple way to set the weights so that the desired patterns become energy minima (attractors) of the network dynamics.
 - d. Hebbian learning strengthens the connection between two neurons that are active at the same time.
4. What does the following figure illustrate? [more than one answer is correct]



- a. The computation of the margin of the decision boundary in linear binary classification through the perceptron learning algorithm.
 - b. The Hopfield network convergence dynamics towards attractors.
 - c. A supervised learning task.
 - d. The adjustment of the decision boundary after each update step.
5. In what scenario is a deep learning model generally preferred over a traditional machine learning model? [more than one answer is correct]
 - a. You are predicting housing prices using structured data including number of rooms, location, and area size, with 3000 training samples.
 - b. You are building a speech recognition system to transcribe recorded conversations into text, with thousands of hours of audio data available but no manually extracted features.
 - c. You are building a sentiment analysis model to determine whether customer reviews are positive or negative. Your dataset consists of 1500 text reviews, each already converted into simple bag-of-words numerical representations.
 - d. You are designing an autonomous driving system that interprets real-time video feeds from multiple cameras to detect pedestrians and other vehicles, using a dataset of millions of labeled images and videos.



6. Given the following 4 x 4 feature map:

$$\begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

If you apply average pooling with a 2 x 2 filter and stride 2, what are the values of the pooled output? [one answer is correct]

- [5, 6, 8, 9]
 - [3, 2.5, 4, 4]
 - [4, 3, 7, 3]
 - None of the above
7. In an RNN, if the tanh activation function were replaced with ReLU, what is a possible consequence? [one answer is correct]
- The hidden states will remain bounded between -1 and 1.
 - The vanishing gradient problem would worsen.
 - The network may suffer from exploding activations due to unbounded ReLU outputs.
 - The output will become a probability between 0 and 1.
8. Which of the following statements about Backpropagation Through Time (BPTT) are correct? [more than one answer is correct]
- During BPTT, the unfolded RNN is treated similarly to a multi-layer neural network.
 - BPTT involves unfolding the RNN in time during the forward pass.
 - BPTT treats the unfolded RNN as a convolutional neural network for training.
 - Weight updates are calculated for each time step and then combined.
9. What is the vanishing gradient problem caused by in RNNs? [more than one answer is correct]
- By the multiplicative nature of the backpropagation algorithm through time.
 - By large enough singular values of the weight matrix.
 - By the forget gate in the case of LSTMs.
 - The use of activation functions with derivatives less than 1.
10. What does the Universal Approximation Theorem state? [more than one answer is correct]
- Suppose we have a function $f(x)$ that we would like to compute within some desired accuracy $\epsilon > 0$. It is guaranteed that by using enough hidden layers we can always find a neural network whose output $g(x)$ satisfies $|g(x) - f(x)| < \epsilon$ for all inputs x .
 - Any continuous function $f: [0, 1]^n \rightarrow [0, 1]$ can be approximated with arbitrary precision by a neural network with at least 1 hidden layer with a finite number of weights.
 - Suppose we have a function $f(x)$ that we would like to compute within some desired accuracy $\epsilon > 0$. It is guaranteed that by using enough hidden neurons we can always find a neural network whose output $g(x)$ satisfies $|g(x) - f(x)| < \epsilon$ for all inputs x .
 - None of the above.



Part II - Open Questions [50 points]

1. (15 points) Answer the following open questions about perceptron networks.

- a) (5) Explain the Hebbian learning algorithm for the Simple Perceptron considering the case where only one output is returned by the network (write down how the training set looks like, how the corresponding output is computed for each pattern, what is the aim, and how weights are updated).

Hebbian learning algorithm for perceptron

As the **computed output values are all independent** from each other, let's consider the case where we have $M=1$ output.

Given a data set $(\vec{x}^1, y^1), (\vec{x}^2, y^2), \dots, (\vec{x}^P, y^P)$, $\vec{x}^p \in \mathbb{R}^d$, $y^p \in \{-1, 1\}$, we denote Perceptron's output by

$$O^p = \text{sgn}(\vec{w}^T \vec{x}^p + b), \quad p = 1, \dots, P.$$

The aim is to identify values of w^i and b in such a way that:

$$O^p = y^p \quad \text{for } p = 1, \dots, P$$

(p = # target patterns)

Learning Rule: Perform the calculation on the whole training set until the weights converge:

$$\vec{w}' = \vec{w} + \Delta \vec{w}$$

$$\Delta w_j = \begin{cases} 2\eta y_p x_j^p & \text{if } O_p \neq y_p \\ 0 & \text{otherwise} \end{cases}$$

where $0 < \eta < 1$ is the **learning rate**, a small constant (e.g., 0.01), and j = index for input nodes.

- b) (5) Starting from the logical table below, prove using the perceptron's output definition that the data points of the eXclusive OR problem are not linearly separable.

x_1	x_2	y
-1	-1	-1
-1	1	1
1	-1	1
1	1	-1

The XOR function is not-linearly separable, in fact:

x_1	x_2	y	$w_1x_1 + w_2x_2 - w_3$	
-1	-1	-1	$-w_1 - w_2 - w_3 < 0$	(a)
-1	1	1	$-w_1 + w_2 - w_3 > 0$	(b)
1	-1	1	$w_1 - w_2 - w_3 > 0$	(c)
1	1	-1	$w_1 + w_2 - w_3 < 0$	(d)

From (d) and (b) it follows

$$w_1 + w_2 < w_3 < -w_1 + w_2$$

Hence, $w_1 < 0$.

From (a) and (c) it follows

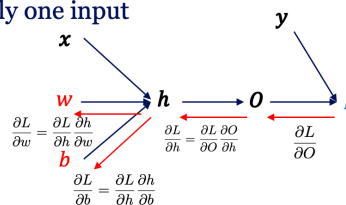
$$-w_1 - w_2 < w_3 < w_1 - w_2$$

From which $w_1 > 0$. **→ CONTRADICTION!**

- c) (5) Consider a single neuron with input x , weight w , bias b , activation function $g(h)$ where $h = wx + b$, and output $O = g(h)$. The neuron is used in a supervised learning setting with a loss function $L(O, y)$, where y is the given output.

- (1) Draw the computation graph,
- (2) derive the expression for the partial derivatives of the loss with respect to the variables of the multi-layer perceptron,
- (3) explicitly define the error term,
- (4) compute the derivative $\frac{\partial O}{\partial h}$ in case g is the ReLu activation function and
- (5) write down δ varies for different values of h ($\geq$ or < 0).

Consider a single node with only one input



The univariate chain rule:

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial O} \frac{\partial O}{\partial h} \frac{\partial h}{\partial w}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial O} \frac{\partial O}{\partial h} \frac{\partial h}{\partial b}$$

$$\frac{\partial h}{\partial w} = x, \frac{\partial h}{\partial b} = 1$$

If we define: $\delta = \frac{\partial L}{\partial O} \frac{\partial O}{\partial h} = \frac{\partial L}{\partial O} g'(h)$

$$\frac{\partial L}{\partial w} = \delta x$$

$$\frac{\partial L}{\partial b} = \delta$$



For ReLU activation:

$$g(h) = \max(0, h)$$

The derivative:

$$\frac{\partial o}{\partial h} = \begin{cases} 1 & \text{if } h > 0 \\ 0 & \text{if } h \leq 0 \end{cases}$$

✓ Impact on δ :

- If $h > 0$, then

$$\delta = \frac{\partial L}{\partial o} \times 1 = \frac{\partial L}{\partial o}.$$

- If $h \leq 0$, then

$$\delta = \frac{\partial L}{\partial o} \times 0 = 0.$$

2. (20 points) Answer the following questions about convolutional neural networks (CNNs).

- a. (4) For what task are majorly used CNNs and why are they preferred to fully connected feed-forward networks (give two reasons)?

Task: Image Classification

Preferred over MLPs because:

- The number of weights (H x W) will explode!
- We destroy the geometric information of the image!

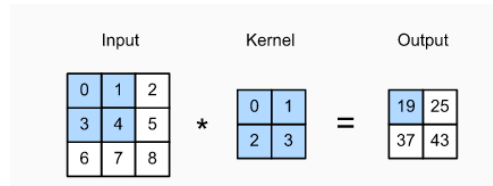
- a. (4) Why is the ReLU activation function commonly used in CNNs?

It introduced non-linearities, enabling CNNs to learn complex patterns and representations.

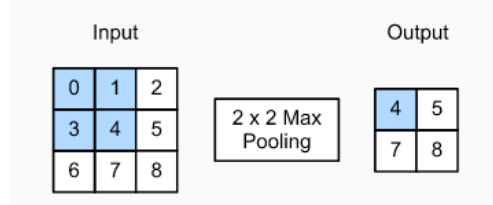
It outputs zeros for negative inputs, leading to some neurons being inactive. This sparsity improves model efficiency and can reduce overfitting.

- b. (6) Explain what convolution and what pooling are. For each of them, provide a visual example.

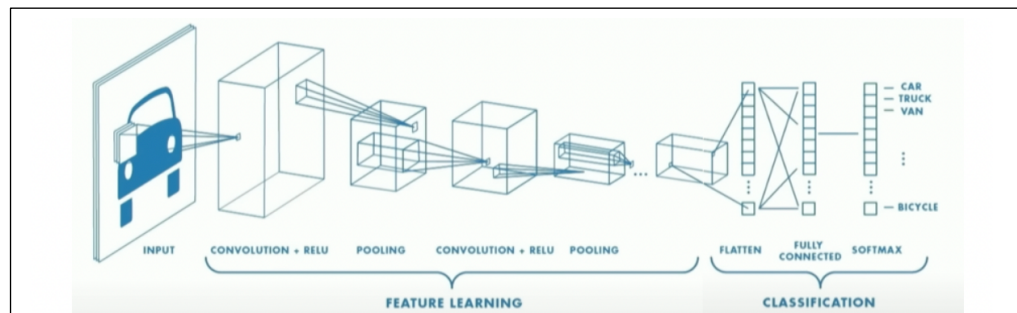
Convolution is an operation used in Convolutional Neural Networks (CNNs) to **extract features** from input data (such as images).



Pooling is an operation used in CNNs to **reduce the spatial dimensions** of feature maps while keeping important information.



- c. (6) Please draw the CNN feature learning + classification pipeline and give a brief description for the two phases.



Feature Learning:

1. Learn features in input images through convolution
2. Introduce non-linearity through activation function (real-world data is nonlinear)
3. Reduce dimensionality and preserve spatial invariance with pooling

Classification:

1. CONV and POOL layers output high-level features of input
2. Fully connected layers uses those features for classifying input images
3. Express output as probability of an image to belong to a particular class

3. (15 points) Answer the following questions about Generative Models:

- a. (6) Please describe the two sub-networks composing an autoencoder and give a mathematical definition for reconstruction error.

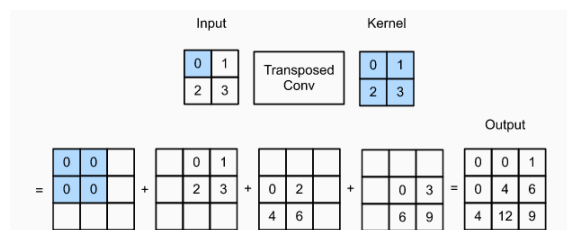
- Two (sub-)networks: encoder and decoder
 - Encoder:** to obtain the latent low-dimensional representation of the data.
 - Decoder:** reconstruct the data from the latent code.

- Loss function - reconstruction error

$$\mathcal{L}_{AE}(\phi, \theta) = \frac{1}{n} \sum_{i=1}^n \left[\vec{x}^{(i)} - f_{\theta} \left(g_{\phi} \left(\vec{x}^{(i)} \right) \right) \right]^2$$

- Mathematically, we try to learn an identity function: $\text{id} \approx f_{\theta} \circ g_{\phi}$

- b. (4) Provide an example to show how transposed convolution works. In what phase of the autoencoder network is it used?



It is used in the decoding phase.

- c. (5) Explain the training process of a Generative Adversarial Network (GAN). In your answer, describe:

- The roles of the generator (G) and discriminator (D).
- How G and D respond based on whether D is correct or wrong.
- What are D and G "rewarded" for.

[Answer field in the following page]



- The process:
 1. G randomly picks a real sample from the training set and alters it with noise to generate a new image.
 2. D receives a mix of real and synthetic images and attempts to distinguish between them.
 3. Based on D's success or failure, both D and G adjust their parameters to improve.
 - If D is correct, G improves to create better fakes.
 - If D is wrong, D learns to avoid similar mistakes.
 4. D is rewarded for correct predictions, while G is rewarded for D's errors.
- This iterative process continues until both networks reach an optimal balance.